

Alonso Marcos Muñoz

Data Engineer | Python | SQL | Spark | Databricks | AWS

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SUMMARY

Data Engineer with almost 2 years of experience in ETL pipelines, PostgreSQL and Oracle modeling and production deployment. Part of the ImpulsaDATA platform team at Tragsatec (15 people in a project of around 80), serving 22 Spanish ministries and public bodies. MSc in Big Data and Cloud Computing with hands-on Databricks, Spark, Kafka, Airflow and AWS.

EXPERIENCE

Tragsatec

Albacete, Spain

Data Engineer

Oct. 2025 - Present

- Build and maintain Python and SQL ETL pipelines that collect, validate and publish ImpulsaDATA catalogs daily or weekly, in a program that has released 5,771 open datasets.
- Design PostgreSQL and Oracle transformations, aggregations and views ready to be queried and published.
- Evolve the relational metadata model with the team, protecting data integrity in every schema change.
- Automate the checks that validate every dataset against the European open data standard, keeping errors out of the official catalogs of 22 public bodies.
- Run development, test, pre-production and production environments with Docker, Podman, Git and SSH.
- Contribute to GitLab CI/CD pipelines that verify code before each deployment and make it repeatable, with automated tests, backups and recovery on failure.
- Within the AgoraData project, lead development of the web interface designed for the data office (CDO, data owners and stewards) to operate ImpulsaDATA's backend and Python processes.

Tragsatec

Albacete, Spain

Data Engineer (Internship)

Jun. 2025 - Sep. 2025

- Developed the first version of the project's metadata conversion tool, a Python ETL that automatically publishes in CKAN public data that was not previously available as open data.
- Added external configuration, execution logs and documentation so the team could maintain the tool.

La Fábrica del Tiempo

Albacete, Spain

Data & Automation Engineer (Internship)

Feb. 2024 - Aug. 2024

- Improved the efficiency of lead management, notifications and approvals by more than 25%, automating them with Power Automate across SharePoint, the CRM and Microsoft 365.
- Delivered an internal Power Apps application that centralizes processes and information for 10+ people.
- Trained the team on the new tools and prepared technical content to support their adoption.

PROJECTS

- **Databricks Telco Churn Lakehouse:** Created an end-to-end data platform that predicts telecom customer churn from 16 million synthetic records of 2 million customers, using PySpark, Delta Lake, Unity Catalog and MLflow: <https://github.com/alonsomarcosm99/databricks-telco-churn-lakehouse>
- **Batch and streaming platform:** Implemented a batch and real-time data platform with Spark, Kafka and Airflow that organizes data into layers of increasing quality (Medallion architecture) on Delta Lake: <https://github.com/alonsomarcosm99/spark-kafka-airflow-data-platform>
- **AWS Smart Parking:** Deployed an AWS IoT platform that delivers data from 40 simulated parking sensors to its API in under 2 seconds at an estimated cost below USD 5 per month, using Lambda, DynamoDB and MQTT: <https://github.com/alonsomarcosm99/smart-parking-albacete>

EDUCATION AND CERTIFICATIONS

UCLM - MSc Big Data and Cloud Computing

2025 - 2026

- Spark, Databricks, streaming, AWS, machine learning, IoT and data governance. Master's thesis graded 9.2/10.

UCLM - BSc Computer Engineering, Information Technology

2019 - 2025

IES Juan Bosco - Higher Diploma in Networked Systems Administration

2017 - 2019

CAPM, project management certification - Project Management Institute

2025

B2 First - Cambridge English

2025

SKILLS

Data Engineering: SQL, Python, ETL and ELT, data modeling, PostgreSQL, Oracle, REST APIs and data quality.

Applied Big Data and cloud: Databricks, Spark, PySpark, Airflow, AWS, Kafka, Delta Lake, MLflow and Terraform.

Development and operations: Git, GitLab CI/CD, Docker, Podman, Linux, SSH, Java, Spring Boot and Vaadin.

Platforms: CKAN, OpenMetadata, DCAT-AP-ES, RDF, JSON-LD, SHACL, Power Apps and Power Automate.

Languages: native Spanish and professional English (B2 First).